

Proposed Solution

Date	30 September 2026
Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Project Overview

Item	FitBuddy
Objective	Provide personalized 7-day workout plans using generative AI and make them adjustable through user feedback.
Scope	Web-based fitness planning, nutrition/recovery tip generation, plan persistence, feedback-based revision and admin viewing.
Problem Statement	Users may struggle to create a structured routine that matches their goals and intensity, and static plans are difficult to adjust.
Impact	Reduces manual planning effort and provides a structured starting point that users can revise through feedback.

Proposed Approach

FitBuddy collects a small fitness profile, sends it to Gemini to generate a structured 7-day plan, creates a concise nutrition/recovery tip, stores the results in SQLite, and uses later feedback to revise the plan while preserving the 7-day structure.

Key Features

Feature	Description
Personalized 7-day plan	Workout focus, warm-up, main workout, sets/repetitions or duration, rest and cool-down.
Nutrition / recovery tip	One concise tip aligned with the selected fitness goal.
Feedback-based revision	User feedback is combined with the original plan to create an updated plan.
Persistent records	User details and plan versions are stored in SQLite.
Admin dashboard	Authenticated admin can view users and associated plans.

Resource Requirements

Type	Requirement
Hardware	A computer capable of running Python and a local web server.
Software	Python, FastAPI, Uvicorn, Jinja2, SQLAlchemy, SQLite and required packages.
API	Google Gemini API key configured through environment variables.
Development	VS Code or another Python-compatible IDE; Git/GitHub for version control.